

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

Zhou Hang

Designing a linear-phase FIR filter using the window method

- Designing Principles
- Designing Methods
- Window Choosing
- Designing Examples

Designing Principles

Designing FIR digital filters using the window method:

Approximate the unit impulse response of an ideal filter in the time domain.

IIR digital filter design is achieved by designing analog filters;

Analog filters are realized by designing three prototype low-pass filters;

Low-pass filters are designed by specifying the cutoff frequencies and their attenuations.

Designing Principles

Time domain approximation concept:

Given $H_d(e^{j\Omega})$,use the Inverse Discrete-Time Fourier Transform (IDTFT) to find its unit impulse response $h_d[k]$.

$$h_d[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(e^{j\Omega})e^{jk\Omega}d\Omega$$

$h_d[k]$ is generally a non-causal, infinite-length sequence.

Windowing and truncating $h_d[k]$, to make it a causal, finite-length sequence $h[k]$, $h[k]$ is a time-domain approximation of $h_d[k]$.

Designing Methods

1. Determine the type and magnitude function $A_d(W)$ of the FIR digital filter from $H_d(e^{jW})$

2. Determine the phase $\varphi_d(W)$ of the linear-phase FIR filter based on its type.

$$\varphi_d(W) = -0.5MW + b \qquad (b = 0 \text{ or } p/2)$$

3. Use $A_d(W)$ and $\varphi_d(W)$ to find $h_d[k]$ through the Inverse Discrete-Time Fourier Transform (IDTFT).

$$h_d[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} A_d(\Omega)e^{j\varphi_d(\Omega)}e^{jk\Omega}d\Omega$$

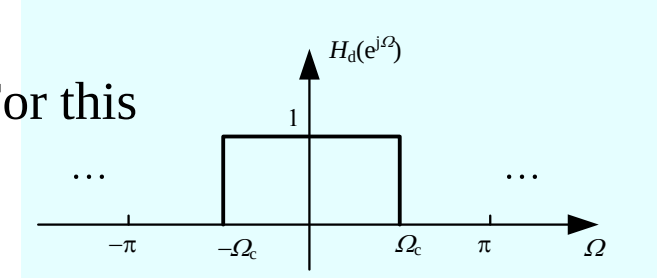
4. Window and truncate $h_d[k]$, to obtain a finite-length causal sequence $h[k]$

$$h[k] = h_d[k]w_N[k]$$

Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=0.5\pi$ rad, for the magnitude response $H_d(e^{j\Omega})$.

Solution: (1) : Determine the type of the linear-phase FIR filter from $H_d(e^{j\Omega})$

A low-pass filter can be of Type I or Type II. For this example, choose Type I.



Determine the magnitude function $A_d(\Omega)$ from $H_d(e^{j\Omega})$:

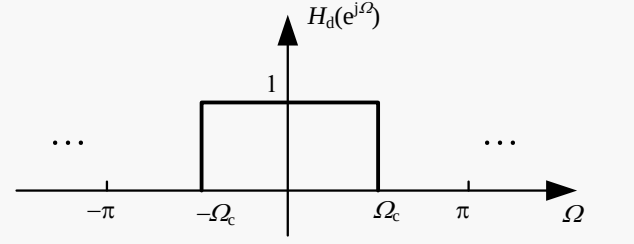
$$A_d(\Omega)=\begin{cases}1 & |\Omega|\le\Omega_c\\ 0 & \text{others}\end{cases}$$

(2) Determine the phase $\varphi_d(\Omega)$ of the linear-phase FIR filter based on its type:

For a Type I linear-phase filter, the phase $\varphi_d(\Omega)$ is: $\varphi_d(\Omega)=-0.5M\Omega$

Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=0.5\pi$ rad, for the magnitude response $H_d(e^{j\Omega})$.

Solution: (3) Determine $h_d[k]$ by applying the Inverse Discrete-Time Fourier Transform (IDTFT) to $A_d(\Omega)$ and $\varphi_d(\Omega)$



$$h_d[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi} A_d(\Omega)e^{j\varphi_d(\Omega)}e^{jk\Omega}d\Omega$$

$$=\frac{1}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{j\Omega(k-0.5M)}d\Omega=\frac{\Omega_c}{\pi}\mathrm{Sa}[\Omega_c(k-0.5M)]$$

(4) Window and truncate $h_d[k]$ to obtain a finite-length causal sequence $h[k]$.

$$h[k]=h_d[k]w_N[k]$$

A window function of length N (N=M+1)

Window Choosing

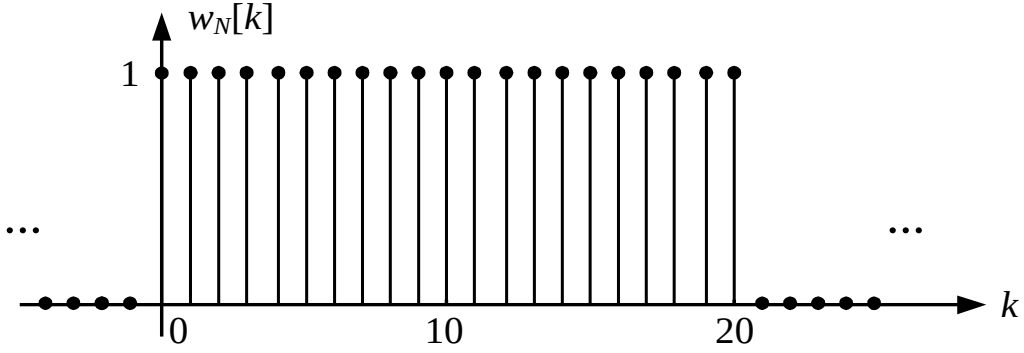
- Rectangular window
- Weighted window
- Adjustable window

Designing a linear-phase FIR filter using the rectangular window method

Rectangular window

$$h[k] = h_{\mathrm{d}}[k] w_N[k]$$

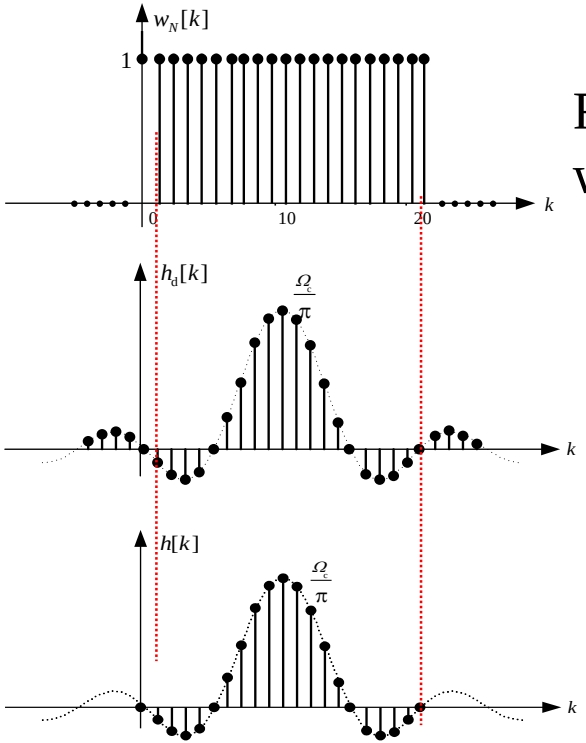
$$w_N[k] = \begin{cases} 1 & 0 \le k \le M \\ 0 & \text{others} \end{cases}$$



Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=0.5\pi$ rad for the magnitude response $H_d(e^{j\Omega})$.

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Use a rectangular window for truncation.



Rectangular window $w_N[k]=\begin{cases}1 & 0\le k\le M\\ 0 & \text{others}\end{cases}$

$$h_d[k]=\frac{\Omega_c}{\pi}\mathrm{Sa}[\Omega_c(k-0.5M)]$$

$$h[k]=h_d[k]\cdot W_N[k]$$

$N=M+1$

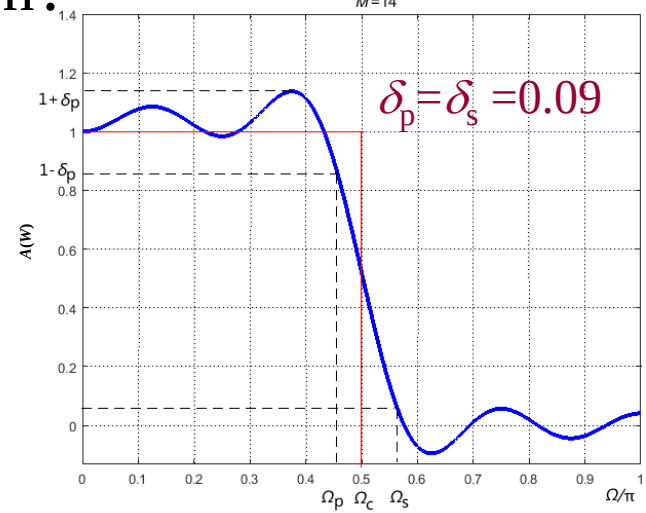
Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=0.5\pi$ rad for the magnitude response $H_d(e^{j\Omega})$.

Use a rectangular window for truncation.

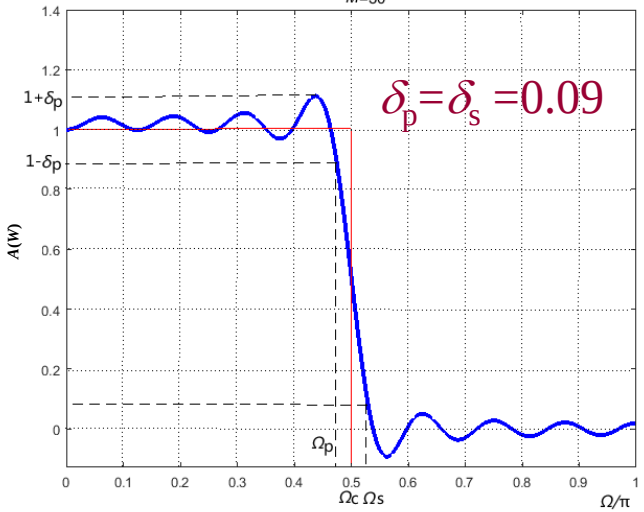
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Solution:

M=14



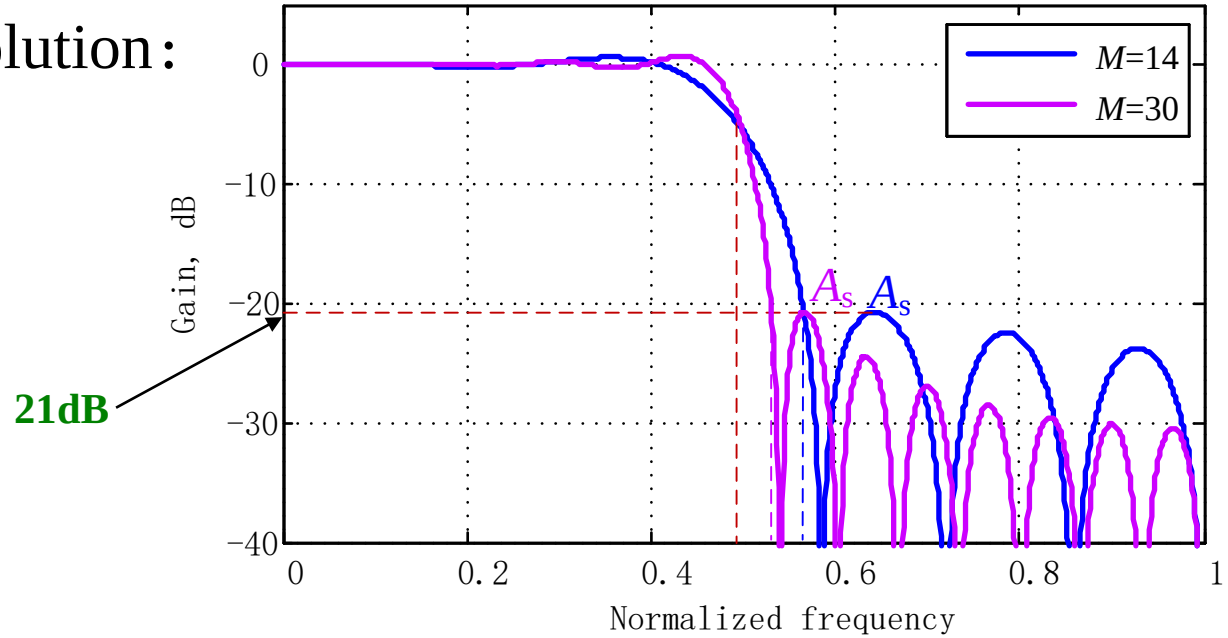
M=30



The designed digital filter's magnitude function exhibits oscillations in both the passband and stopband, with the maximum ripple being approximately 9% of the amplitude transition value. This phenomenon is known as the Gibbs phenomenon.

Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=0.5\pi$ rad for the magnitude response $H_d(e^{j\Omega})$. Use a rectangular window for truncation.

Solution:



Due to the presence of the Gibbs phenomenon

$\delta_p=\delta_s=0.09$

Therefore, the filter order increases
The stopband attenuation remains unchanged.
But the transition zone decreases.

The width of the transition zone is approximately:

$1.8\pi/N$

$A_p=-20\lg(1-\delta_p)\approx 0.82\text{dB}$ $A_s=-20\lg\delta_s\approx 21\text{dB}$

Designing a linear-phase FIR filter using the rectangular window method

Design results analysis

$$h[k] = h_{\mathrm{d}}[k]w_N[k] \qquad N=M+1$$

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{2\pi} H_{\mathrm{d}}(\mathrm{e}^{\mathrm{j}\Omega}) * W_N(\mathrm{e}^{\mathrm{j}\Omega})$$

Window truncation

Using the properties of the DTFT, the magnitude function $A(\Omega)$ of the designed FIR filter can be obtained.

$$A(\Omega) == \frac{1}{2\pi} A_{\mathrm{d}}(\Omega) * W_N(\Omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} A_{\mathrm{d}}(\theta) W_N(\Omega - \theta) \mathrm{d}\theta$$

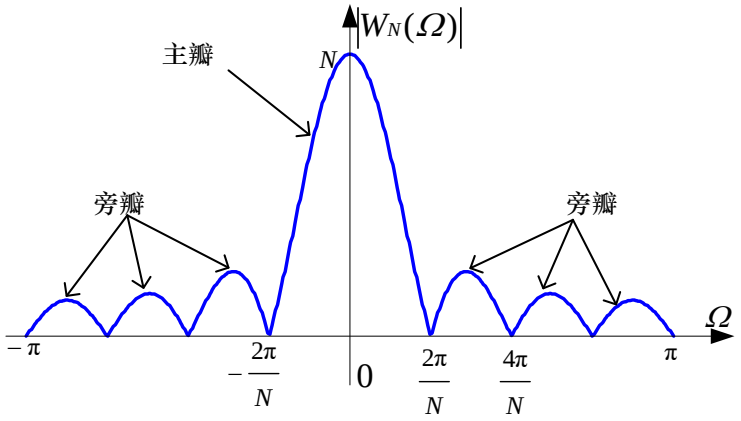
The accuracy of $A(\Omega)$ in approximating $A_{\mathrm{d}}(\Omega)$ depends on the magnitude spectrum $W_N(\Omega)$ of the window function.

Designing a linear-phase FIR filter using the rectangular window method

The frequency spectrum of the rectangular window function.

$$W_N(e^{j\Omega}) = e^{-j\Omega(N-1)/2}\frac{\sin(N\Omega/2)}{\sin(\Omega/2)}$$

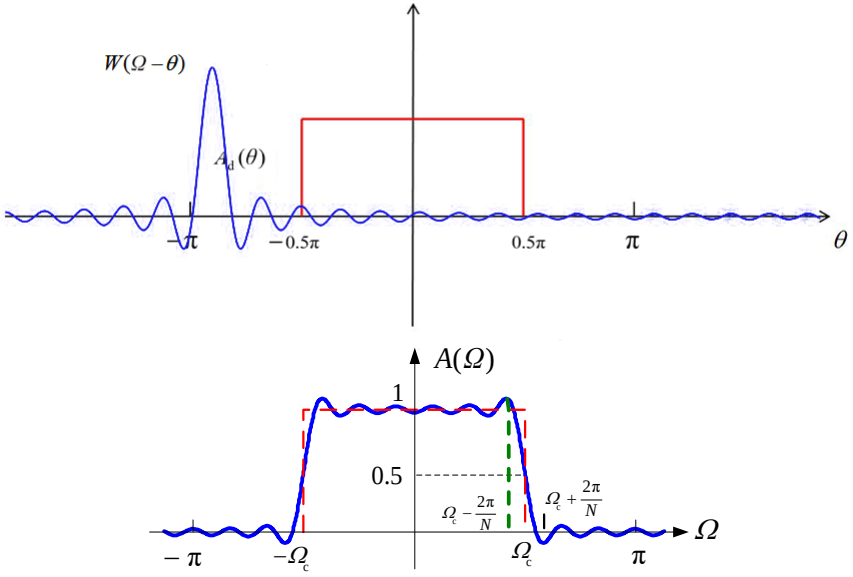
The magnitude function $W_N(\Omega)$ of the rectangular window



1. The main lobe width of $W_N(\Omega)$ is $4\pi/N$
2. The relative attenuation of the side lobes is constant.

Designing a linear-phase FIR filter using the rectangular window method

$$A(\Omega)=\frac{1}{2\pi}A_{\mathrm{d}}(\Omega)*W_N(\Omega)$$



1. The main lobe width of the window function determines the width of the transition band of $H(e^{j\Omega})$. As the window function length N increases, the transition band decreases.

2. The size of the side lobes determines the attenuation of the FIR filter in the stopband.

3. The maximum stopband attenuation of a filter designed using a rectangular window is approximately: $-20\lg(9\%)\approx 21\mathrm{dB}$

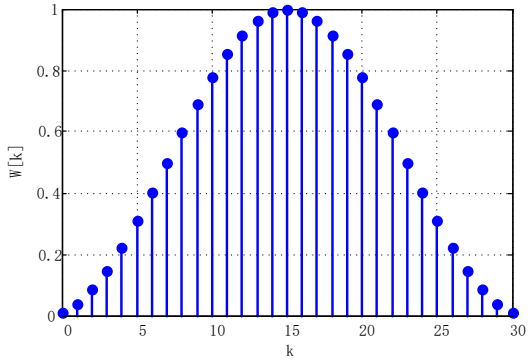
How to improve stopband attenuation?

Weighted window

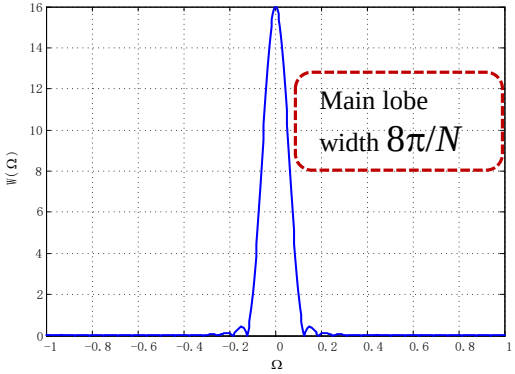
Hanning window (*w*=hanning(*N*))

$$h[k]=h_d[k]w_N[k]$$

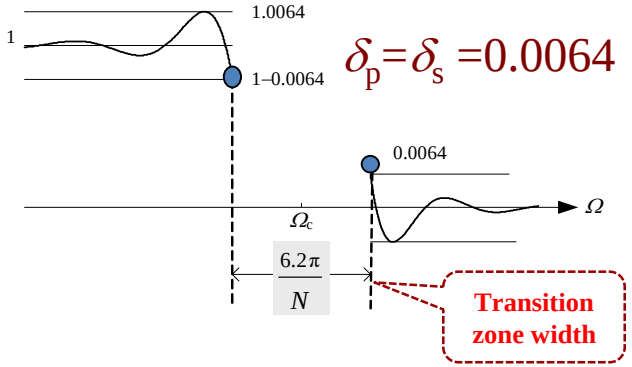
$$w_N[k]=[0.5-0.5\cos(\frac{2\pi k}{N-1})]R_N[k]$$



Time-domain waveforms （*M*=30）



Spectrum of amplitude of the window function (*M*=30)



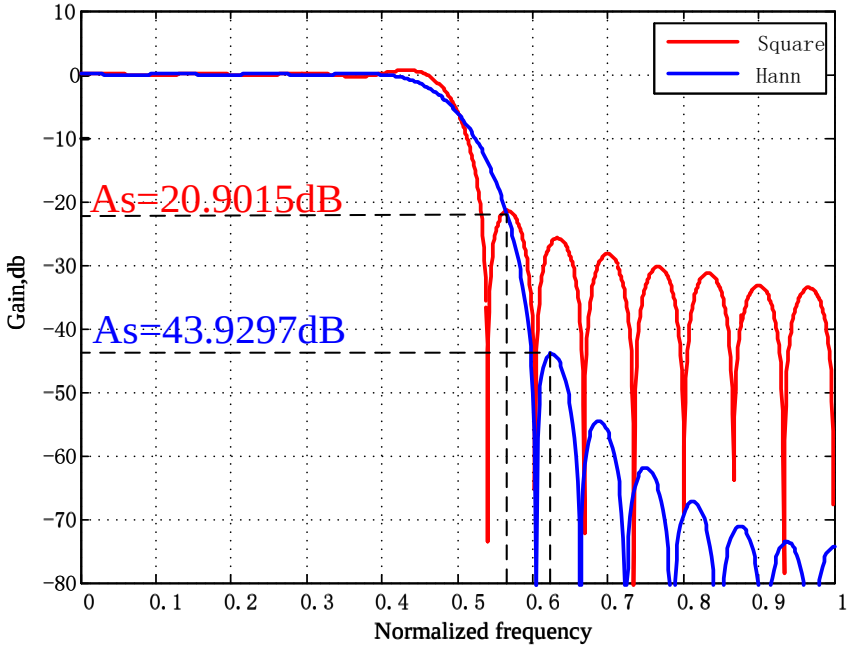
Filter amplitude function

$A_p \approx$ 0.056dB, $A_s \approx$ 44dB

Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=\pi/2$ rad for the magnitude response $H_d(e^{j\Omega})$. Use Hann window, Hamming window, and Blackman window for truncation, and compare the results with those from a rectangular window design.

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Solution: Hann window compared to rectangular window ($\Omega_c=\pi/2$, $M=30$)

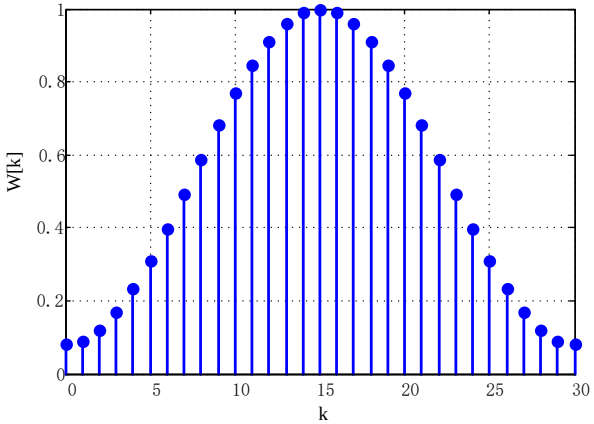


Weighted window

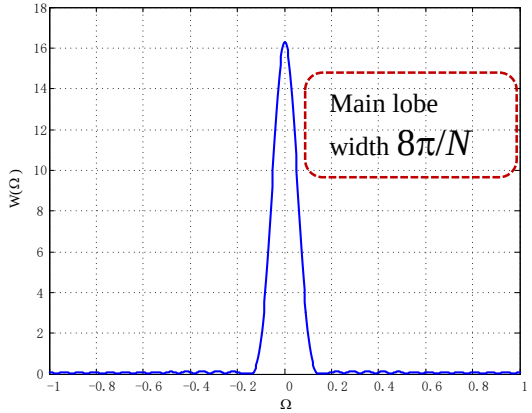
Hamming window (*w*=hamming(*N*))

$$h[k]=h_{\mathrm{d}}[k]w_N[k]$$

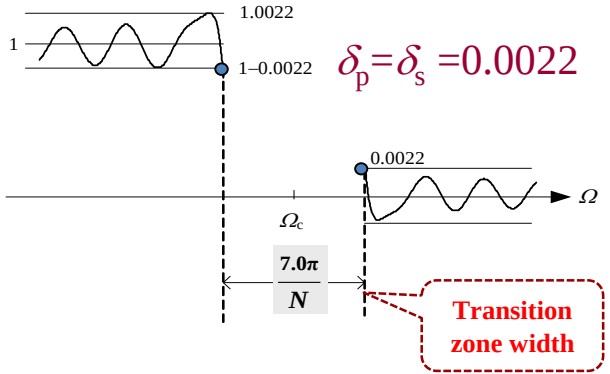
$$w_N[k]=[0.54-0.46\cos(\frac{2\pi k}{N-1})]R_N[k]$$



Time-domain waveforms （*M*=30）



Spectrum of amplitude of the window function （*M*=30）

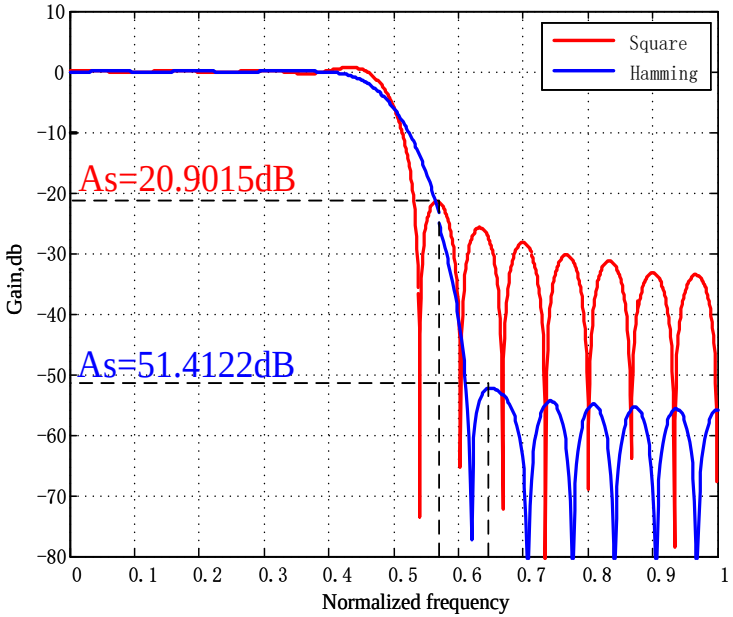


Filter amplitude function

$A_p \approx$ 0.019dB, $A_s \approx$ 53dB

Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=\pi/2$ rad for the magnitude response $H_d(e^{j\Omega})$. Use Hann window, Hamming window, and Blackman window for truncation, and compare the results with those from a rectangular window design.

Solution: Hamming window compared to rectangular window ($\Omega_c=\pi/2$, $M=30$)

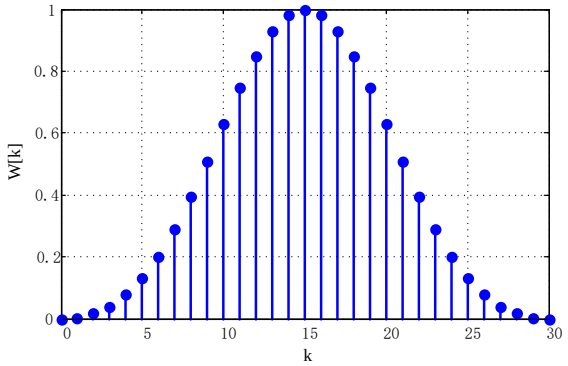


Weighted window

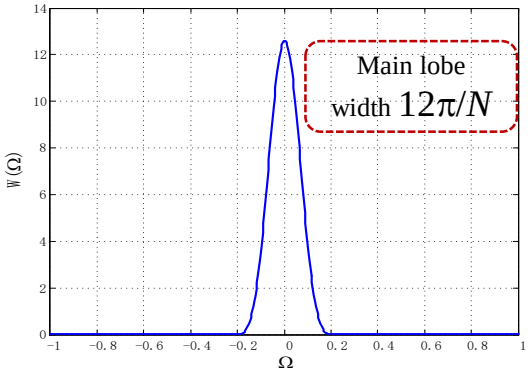
Blackman window (*w*=blackman(*N*))

$$h[k]=h_d[k]w_N[k]$$

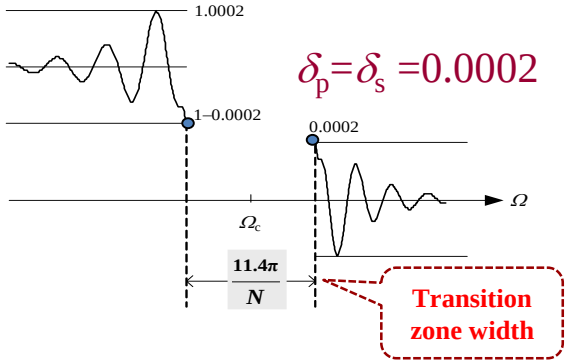
$$w_N[k]=[0.42-0.5\cos(\frac{2\pi k}{N-1})+0.08\cos(\frac{4\pi k}{N-1})]R_N[k]$$



Time-domain waveforms （*M*=30）



Spectrum of amplitude of the window function (*M*=30)



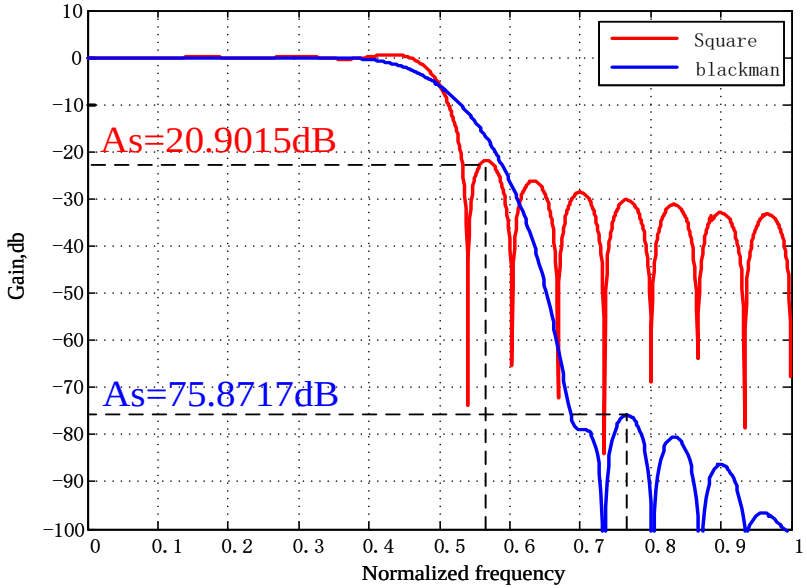
Filter amplitude function

$A_p \approx$ 0.0017dB，$A_s \approx$ 74dB

Example: Design a linear-phase FIR filter using the window method to approximate a low-pass filter with a cutoff frequency $\Omega_c=\pi/2$ rad for the magnitude response $H_d(e^{j\Omega})$. Use Hann window, Hamming window, and Blackman window for truncation, and compare the results with those from a rectangular window design.

Solution:

Blackman window compared to rectangular window ($\Omega_c=\pi/2$, $M=30$)



Weighted window

Properties of commonly used window functions

Type	Mainlope width	approximately transition bandwidth	δ_p, δ_s	A_p(dB)	A_s(dB)
Rectangular	$4\pi / N$	$1.8\pi / N$	0.09	0.82	21
Hann	$8\pi / N$	$6.2\pi/N$	0.0064	0.056	44
Hamming	$8\pi / N$	$7.0\pi/N$	0.0022	0.019	53
Blackman	$12\pi / N$	$11.4\pi/N$	0.0002	0.0017	74

Example: Design a Type I linear-phase FIR low-pass filter using the window method to meet the following specifications.

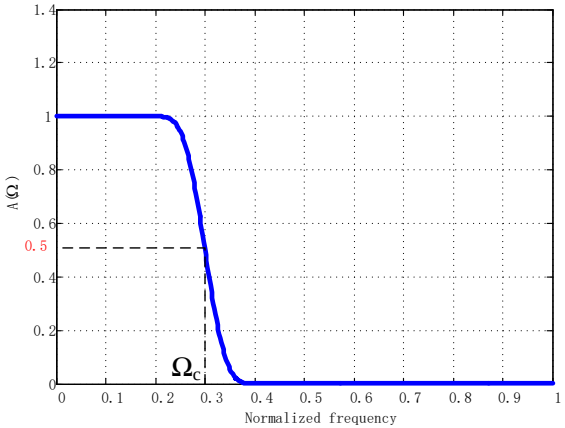
Ω_p=0.2p rad, Ω_s=0.4p rad， A_p=0.3dB, A_s=55dB

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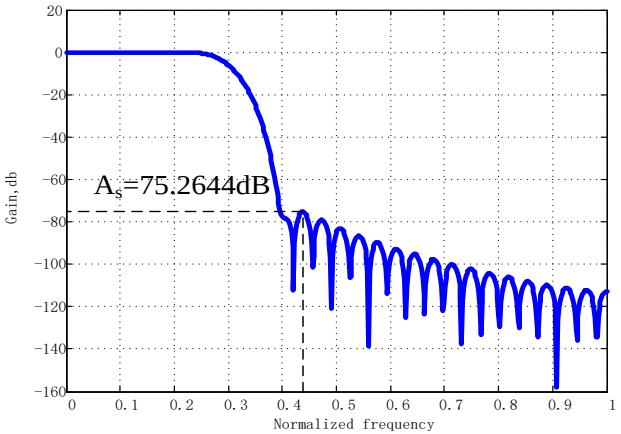
Analysis: Based on the design specification of stopband attenuation A_s=55dB, only the Blackman window can meet the design requirements.

The order $M=N-1$ is determined by the window type and the transition band width.

$11.4p/N=\Omega_s-\Omega_p$ N=57 M=56



Ω_c= (Ω_p + Ω_s)/2=0.3π rad

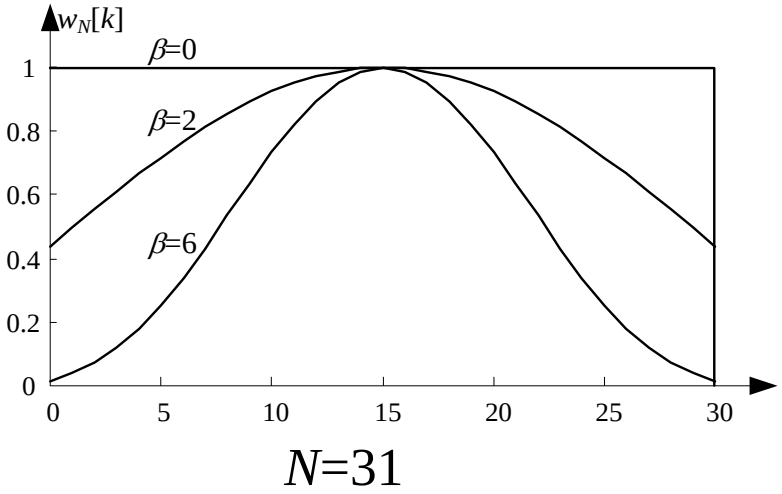


A_s=75.3dB

Adjustable window

Kaiser window (*w*=kaiser(*N*, beta))

$$h[k] = h_d[k] w_N[k] \qquad w_N[k] = \frac{I_0(\beta\sqrt{1-[1-2k/M]^2})}{I_0(\beta)}, \qquad 0 \le k \le N-1$$



N=31

β : Adjustable parameters to modify the shape of the window function

$I_0(x)$:
The modified zeroth-order Bessel function

Adjustable window

Determination of M and β

$$A = -20\lg\left(\min\{\delta_p,\ \delta_s\}\right)$$

$$\beta = \begin{cases} 0.1102(A-8.7), & A > 50 \\ 0.5842(A-21)^{0.4} + 0.07886(A-21), & 21 \le A \le 50 \\ 0, & A < 21 \end{cases}$$

$$M \approx \frac{A-7.95}{2.285\left|\Omega_p - \Omega_s\right|}, \qquad A > 21$$

Example: Design a Type I linear-phase FIR low-pass filter using the Kaiser window to meet the following specifications.

Ω_p=0.2π rad，Ω_s=0.4π rad，A_p=0.3dB，A_s=55dB

Solution:

1. Determine the cutoff frequency of the low-pass filter.

$$\Omega_c = (\Omega_p + \Omega_s)/2 = 0.3\pi \text{ rad}$$

2. Calculate $h_d[k]$ of the ideal linear-phase low-pass digital filter.

$$h_d[k] = 0.3\mathrm{Sa}\left(0.3\pi(k - 0.5M)\right)$$

3. Determine the parameters ***b*** and the order ***M*** of the Kaiser window based on passband and stopband ripples and the transition band.

A= -20lg (min{d_p，d_s })=A_s

b = 0.1102(A_s-8.7)=5.1

$$M \approx \frac{A_s - 7.95}{2.285\left|\Omega_p - \Omega_s\right|} = 32.77$$

M=34

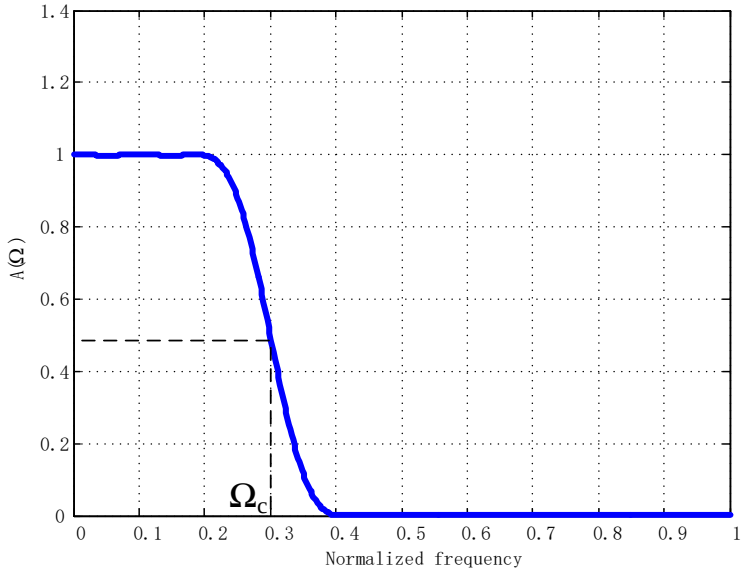
4. $h[k]=h_d[k]\cdot w_N[k]$

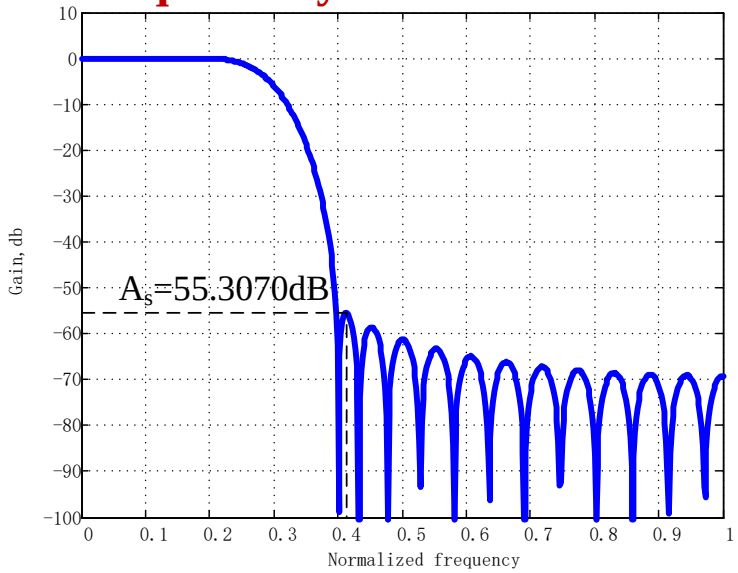
Example: Design a Type I linear-phase FIR low-pass filter using the Kaiser window to meet the following specifications.

Ω_p=0.2π rad， Ω_s=0.4π rad， A_p=0.3dB， A_s=55dB

Solution： **Design result** M=34 A_s=55.307 dB

Achieve the same technical specifications with a lower order than required by the Blackman window.

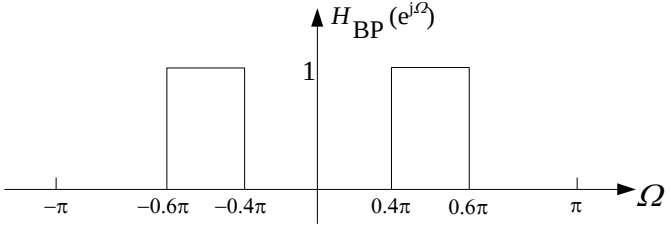




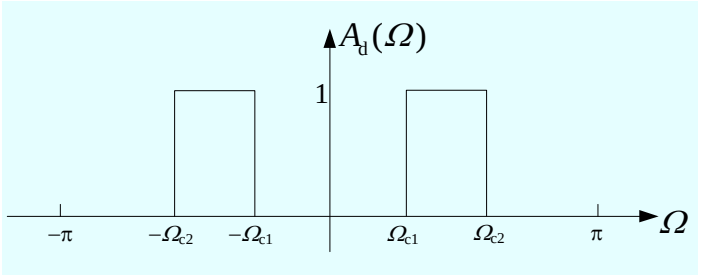
Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

Solution: (1) Determine the type of linear-phase FIR filter, choose Type I.

Determine the magnitude function $A_d(\Omega)$ of the ideal filter, $A_d(\Omega)$ is even symmetric about $\Omega=0$.



$$A_d(\Omega)=\begin{cases}1 & \Omega_{c1}\le|\Omega|\le\Omega_{c2}\le\pi\\ 0 & \text{others}\end{cases}$$



(2) Determine $\varphi_d(\Omega)$, the phase of the ideal filter based on its type.

$\varphi_d(\Omega)=-0.5M\Omega$ M is even

Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

Solution: (3) Calculate IDFT to obtain $h_d[k]$

$$h_d[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi} A_d(\Omega)e^{j\varphi_d(\Omega)}e^{jk\Omega}d\Omega$$

$$=\frac{1}{2\pi}\int_{-\Omega_{c2}}^{-\Omega_{c1}} e^{j\Omega(k-0.5M)}d\Omega+\frac{1}{2\pi}\int_{\Omega_{c1}}^{\Omega_{c2}} e^{j\Omega(k-0.5M)}d\Omega$$

$$=\frac{\Omega_{c2}}{\pi}\mathrm{Sa}[\Omega_{c2}(k-0.5M)]-\frac{\Omega_{c1}}{\pi}\mathrm{Sa}[\Omega_{c1}(k-0.5M)]$$

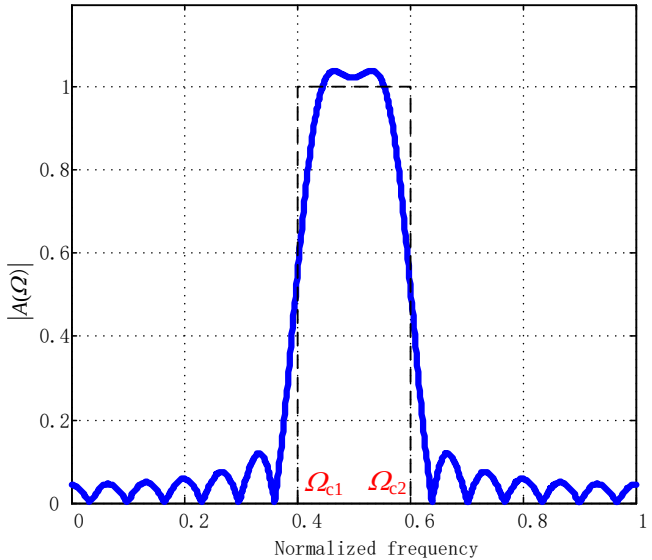
(4) Window and truncate $h_d[k]$ to obtain $h[k]=h_d[k]w_N[k]$

Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

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Solution: Use Type I and implement with a rectangular window: $M=30$

$$h[k]=\frac{\Omega_{c2}}{\pi}\mathrm{Sa}[\Omega_{c2}(k-0.5M)]-\frac{\Omega_{c1}}{\pi}\mathrm{Sa}[\Omega_{c1}(k-0.5M)] \qquad k=0,\ldots,30.$$

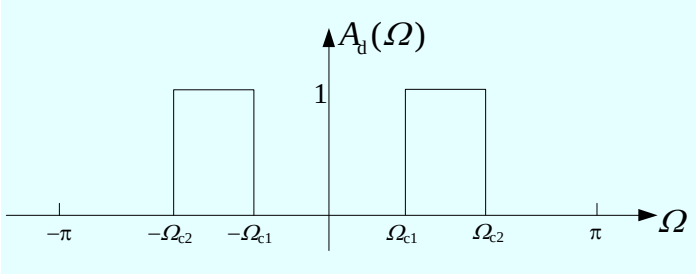


Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

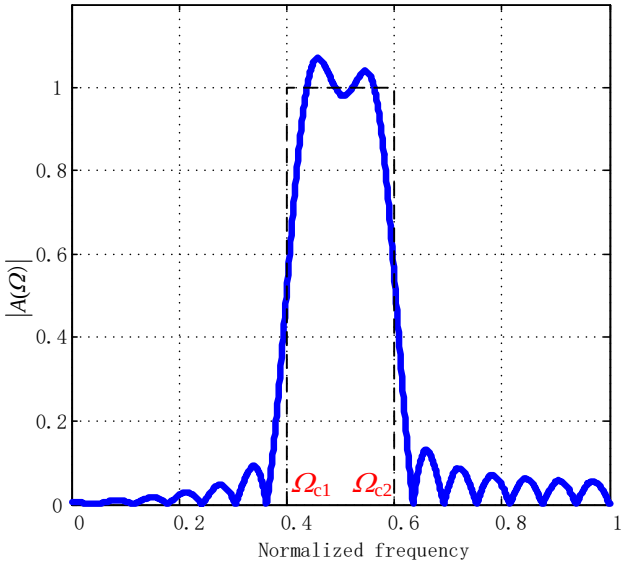
Solution: Use Type II and implement with a rectangular window: $M=31$

$$h[k]=\frac{\Omega_{c2}}{\pi}\mathrm{Sa}[\Omega_{c2}(k-0.5M)]-\frac{\Omega_{c1}}{\pi}\mathrm{Sa}[\Omega_{c1}(k-0.5M)] \quad k=0,\ldots,31.$$

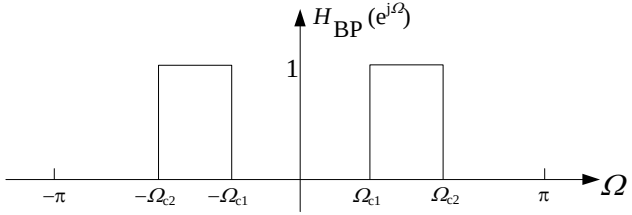
Type II, $A_d(\Omega)$ is even symmetric about $\Omega=0$



$$\varphi_d(\Omega)=-0.5M\Omega$$



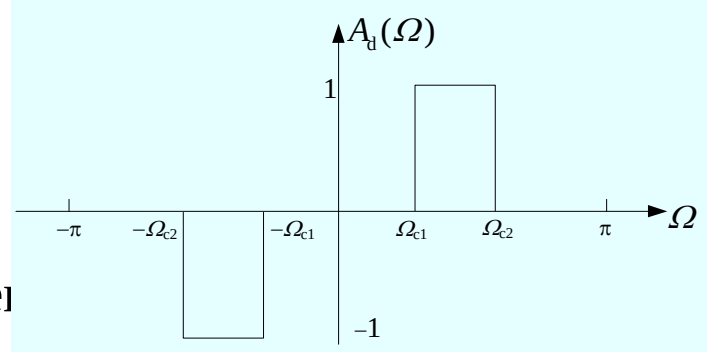
Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.



Solution: (1) Tpye III

Determine the magnitude function $A_d(\Omega)$ of the ideal filter, where $A_d(\Omega)$ is odd symmetric about $\Omega=0$

$$A_d(\Omega)=\begin{cases}1 & 0<\Omega_{c1}\le\Omega\le\Omega_{c2}\le\pi\\ -1 & -\pi\le-\Omega_{c2}\le\Omega\le-\Omega_{c1}<0\\ 0 & \text{others}\end{cases}$$



(2) Determine the phase $\varphi_d(\Omega)$ of the ideal filter based on its type.

$\varphi_d(\Omega)=-0.5M\Omega+\pi/2$ *M is even*

Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

Solution: (3) Calculate the IDTFT to obtain $h_d[k]$

$$h_d[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi} A_d(\Omega)e^{j\varphi_d(\Omega)}e^{jk\Omega}d\Omega$$

$$=\frac{-j}{2\pi}\int_{-\Omega_{c2}}^{-\Omega_{c1}} e^{j\Omega(k-0.5M)}d\Omega+\frac{j}{2\pi}\int_{\Omega_{c1}}^{\Omega_{c2}} e^{j\Omega(k-0.5M)}d\Omega$$

$$=\begin{cases}\dfrac{1}{\pi(k-0.5M)}\{\cos[\Omega_{c2}(k-0.5M)]-\cos[\Omega_{c1}(k-0.5M)]\}, & k\neq 0.5M\\ 0, & k=0.5M\end{cases}$$

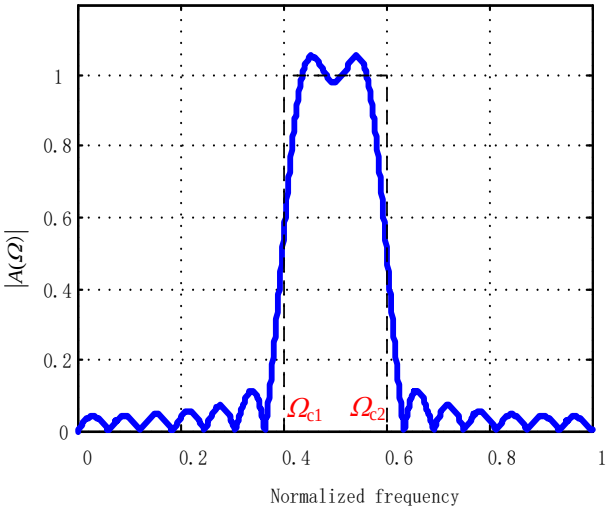
(4) Window and truncate $h_d[k]$ to obtain $h[k]=h_d[k]w_N[k]$

Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

Solution: Use Type III and implement with a rectangular window: $M=30$

$$h[k]=\begin{cases}\dfrac{1}{\pi(k-0.5M)}\{\cos[\Omega_{c2}(k-0.5M)]-\cos[\Omega_{c1}(k-0.5M)]\}, & k\neq 0.5M \\ 0, & k=0.5M\end{cases}$$

$k=0,\ldots,30.$



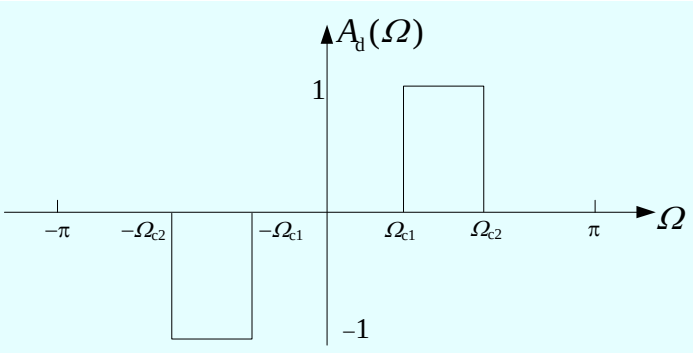
Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$

with $\Omega_{c1}=0.4\pi$ rad，$\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

Solution：Tpye IV

$$h_d[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi} A_d(\Omega)e^{j\varphi_d(\Omega)}e^{jk\Omega}d\Omega$$

$A_d(\Omega)$ is odd symmetric about Ω=0



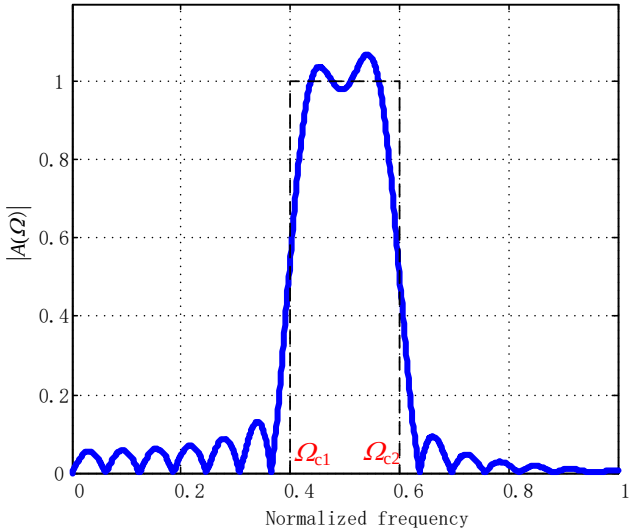
$$=\frac{-j}{2\pi}\int_{-\Omega_{c2}}^{-\Omega_{c1}} e^{j\Omega(k-0.5M)}d\Omega+\frac{j}{2\pi}\int_{\Omega_{c1}}^{\Omega_{c2}} e^{j\Omega(k-0.5M)}d\Omega$$

$$=\frac{1}{\pi(k-0.5M)}\{\cos[\Omega_{c2}(k-0.5M)]-\cos[\Omega_{c1}(k-0.5M)]\}$$

$\varphi_d(\Omega)=-0.5M\Omega+\pi/2$ M *is odd*

Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

Solution: Use Type III and implement with a rectangular window: $M=31$

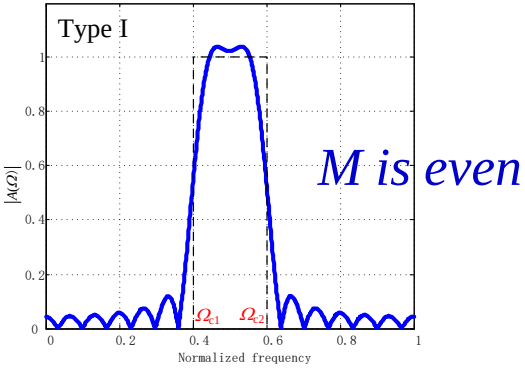


Example: Design a linear-phase FIR filter using the window method to approximate the given band-pass filter $H_{BP}(e^{j\Omega})$ with $\Omega_{c1}=0.4\pi$ rad, $\Omega_{c2}=0.6\pi$ rad, using a rectangular window.

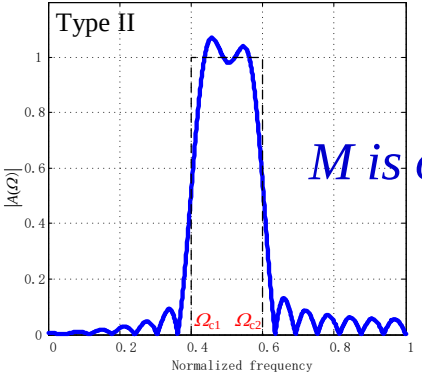
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$$A_d(\Omega)=\begin{cases}1 & \Omega_{c1}\le|\Omega|\le\Omega_{c2}\le\pi\\ 0 & \text{others}\end{cases}$$

$$\varphi_d(\Omega)=-0.5M\Omega$$



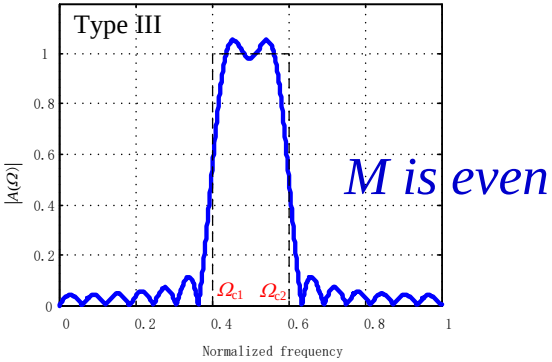
M is even



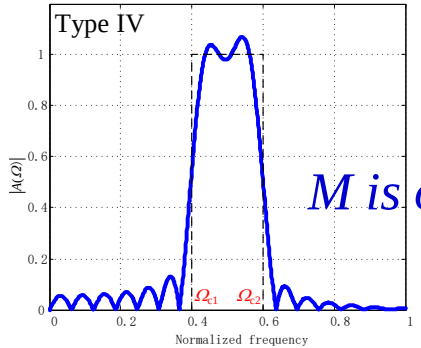
M is odd

$$A_d(\Omega)=\begin{cases}1 & 0<\Omega_{c1}\le\Omega\le\Omega_{c2}\le\pi\\ -1 & -\pi\le-\Omega_{c2}\le\Omega\le-\Omega_{c1}<0\\ 0 & \text{others}\end{cases}$$

$$\varphi_d(\Omega)=-0.5M\Omega+\pi/2$$



M is even



M is odd

Example: Design an FIR high-pass digital filter with the following specifications $\Omega_p=0.6\pi$ rad, $\Omega_s=0.4\pi$ rad， $A_p=0.3$dB, $A_s=40$dB

Solution：1.Since it's a high-pass digital filter, choose Type I design. Then, determine the window function type based on $A_s=40$dB.Options that meet $A_s=40$dB dB include Hann, Hamming, and Blackman windows. The Hann window has a stopband attenuation of 44 dB, which is closest to 40 dB, so the Hann window is selected. $\text{hann}_N[k]$

2. Determine N based Ω_p and Ω_s $\quad N \geq \dfrac{6.2\pi}{|\Omega_s-\Omega_p|}=\dfrac{6.2\pi}{0.2\pi}=31, M=30$

3. $A_d(\Omega)=\begin{cases}1 & |\Omega|>\Omega_c \\ 0 & \text{others}\end{cases}$ mostly $\Omega_c=\frac{1}{2}(\Omega_p+\Omega_s)$ $= 0.5\pi$ rad

$\varphi_d(\Omega)= -0.5M\Omega$

4. $h_d[k]=\delta[k-0.5M]-\dfrac{\Omega_c}{\pi}\text{Sa}[\Omega_c(k-0.5M)]=\delta[k-15]-0.5\text{Sa}[0.5\pi(k-15)]$

5. $h[k]=h_d[k]\cdot\text{hann}_N[k]$ $\quad N=31$

Common window functions and their MATLAB implementation

When designing FIR filters using the window method, $h_d[k]$ needs to be windowed and truncated to obtain a finite-length sequence $h[k]$

$$h[k]= h_d[k]\ w_N[k]$$

Common $w_N[k]$:

- Hanning
- Hamming
- Blackman
- Kaiser

MATLAB:

```
w=hanning(N)
w=hamming(N)
w=blackman(N)
w=kaiser(N,beta)
```

N: length

beta: parameter that control the shape of Kaiser

w: A column vector of length *N* that provides the values of the window function at *N* points.

Example: Design an FIR high-pass digital filter with the following specifications Ω_p=0.6π rad, Ω_s=0.4π rad，A_p=0.3dB, A_s=40dB

The Hann window can be selected for design based on the stopband attenuation specification A_s =40dB.

Key function： N=ceil(6.2/(Wp-Ws)); Find the filter length N

$W_c=(W_p+W_s)/2$; find the cutoff frequency

w=hann(N) ; design hann window

hd=-(Wc/pi)*sinc(Wc*(k-0.5*M)/pi);

hd(0.5*M+1)=hd(0.5*M+1)+1; Ideal high-pass filter unit impulse response

h=hd.*w; The unit impulse response after window addition can be obtained

mag=freqz(h,[1],omega); Find the frequency response characteristic

Example: Design an FIR high-pass digital filter with the following specifications Ω_p=0.6π rad, Ω_s=0.4π rad， A_p=0.3dB, A_s=40dB

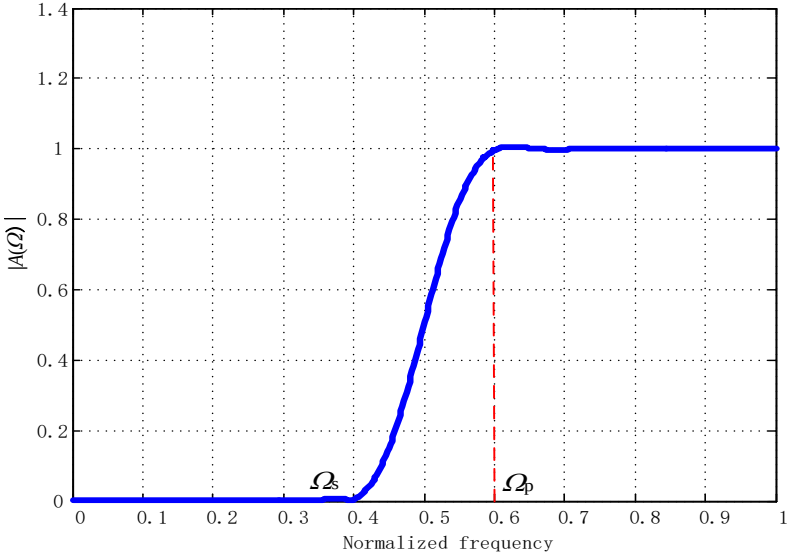
Solution:

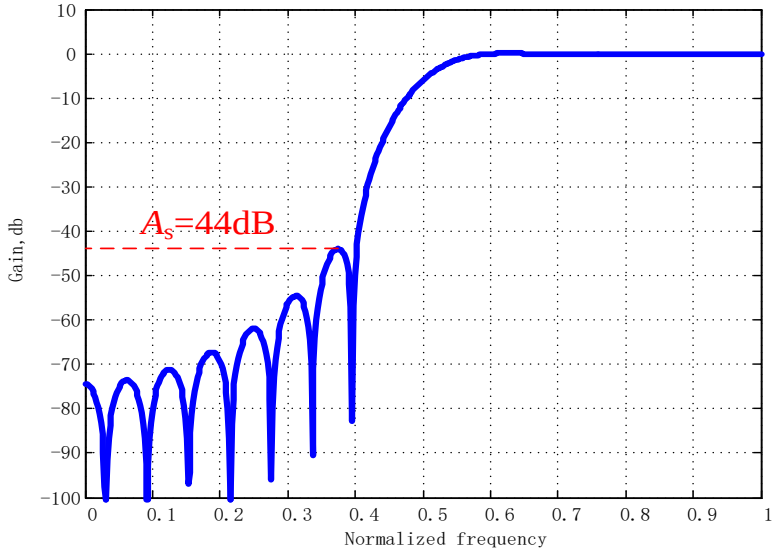
```
% Design FIR high-pass filter using Hann window
Wp=0.6*pi;Ws=0.4*pi;Ap=0.3;As=40;
N=ceil(6.2*pi/(Wp-Ws));
% Determine filter order and ensure it's type I
N=mod(N+1,2)+N;
M=N-1  % Filter order
Mw=hanning(N); % Use Hann window
Wc=(Wp+Ws)/2;
% Ideal high-pass cutoff frequency
k=0:M;
hd=-(Wc/pi)*sinc(Wc*(k-0.5*M)/pi);
hd(0.5*M+1)=hd(0.5*M+1)+1;
h=hd.*w;
omega=linspace(0,pi,512);
% Generate an array from 0 to π
containing 512 points
mag=freqz(h,[1],omega); % Calculate frequency response
% Plot gain response
plot(omega/pi, 20*log10(abs(mag)))
xlabel('Normalized frequency');
ylabel('Gain,db'); figure;
% Plot amplitude response
plot(omega/pi, (abs(mag)),'linewidth',3);grid on
xlabel('Normalized frequency');
ylabel('A(\Omega)');
thet2=unwrap(angle(mag)); % Calculate phase response
figure;
% Plot phase response
plot(omega/pi,thet2,'LineWidth',2); grid on;
xlabel('Normalized frequency');
ylabel('Phase Frequency');grid on
```

Example: Design an FIR high-pass digital filter with the following specifications Ω_p=0.6π rad, Ω_s=0.4π rad， $A_p \le 0.3$dB, $A_s \ge 40$dB

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Solution:





M= 30, A_p=0.055dB, A_s=44dB

MATLAB implementation of window method design

h = fir1(M,Wc,'ftype',window)

M represents the order of the filter.

Wc represents the frequency bands of the ideal FIR filter.
The range of Wc is between 0 and 1.

ftype represents the type of filter, with the default value being empty.

ftype:
- empty (low-pass) :
- 'high' (high-pass) :
- 'stop' (band -stop) :
- 'DC–0' (The first band of the multiband filter is a stopband.)
- 'DC–1' (The first band of the multiband filter is a passband.)

Window is a window function of length $M+1$
If no window function is chosen at the time of calling, the Hamming window is used by default.

h is the $h[k]$ of the designed FIR filter.

Example: Design an FIR high-pass digital filter with the following specifications Ω_p=0.8π rad, Ω_s=0.7π rad， $A_p \leq 0.3$dB, $A_s \geq 40$dB

Key function：

h1=fir1(M,wp0, 'ftype',window); Filter design using the window method

Analysis approach::

For a high-pass filter, set， ‘ftype’ to 'high'

A_s=40， select the Hann window， set window to hanning(N)

Given that the approximate transition bandwidth for the Hann window is (6.2*π/N), the filter length *N* can be calculated using *N*： 6.2*π/N=(Wp-Ws);

The filter order *M*=*N*–1

The cutoff frequency is typically set to *Wc*=*Wp*+*Ws*2W_c=(W_p +W_s)/2， and select a passband W_{p0}=W_c/ π

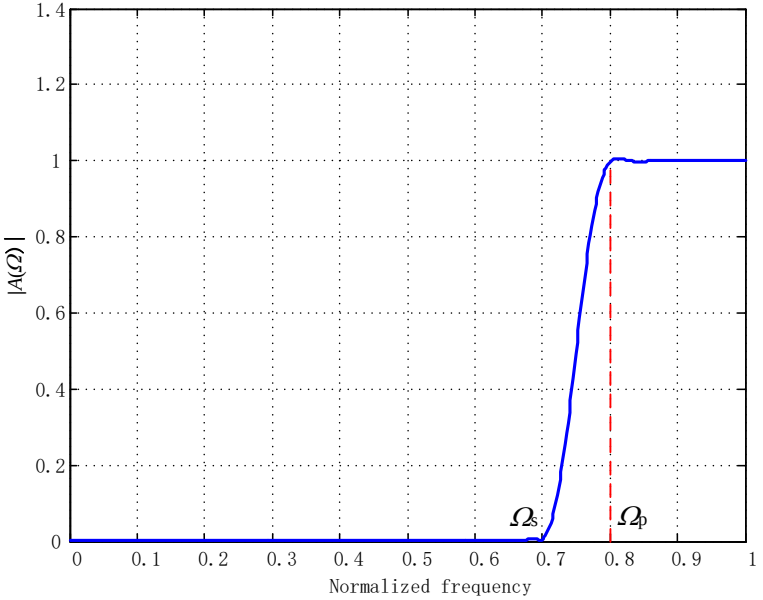
Example: Design an FIR high-pass digital filter with the following specifications $\Omega_p=0.8\pi$ rad, $\Omega_s=0.7\pi$ rad， $A_p \le 0.3$dB, $A_s \ge 40$dB

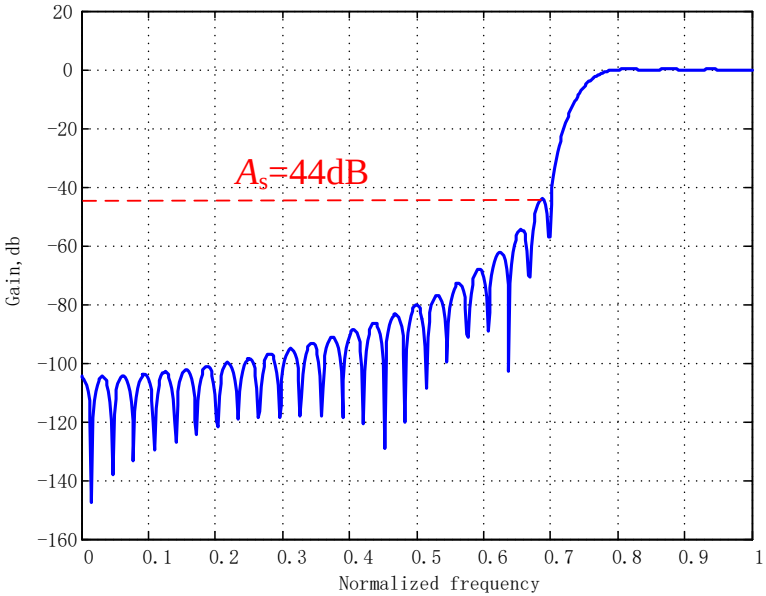
Solution:

```
%fir1design high-pass filter
Wp=0.8*pi; Ws=0.7*pi; Ap=0.3;  As=40;
%Determine filter order and ensure it's type I
N=ceil(6.2*pi/(Wp-Ws));
N=mod(N+1,2)+N;
M=N-1 % Filter order
wp0=(Wp+Ws)/2/pi; % Ideal high-pass cutoff frequency
w=hanning(N);
h1=fir1(M,wp0, 'high',w); % high-pass filter
omega=linspace(0,pi,512);
mag=freqz(h1,1,omega);  %  Calculate frequency response
figure;
% Plot gain response
plot(omega/pi,
20*log10(abs(mag)),'linewidth',2);grid on;
xlabel('Normalized frequency');ylabel('Gain,db');
thet2=unwrap(angle(mag)); % Calculate phase response
figure;
% Plot amplitude response
plot(omega/pi,abs(mag),'LineWidth',2);grid on;
xlabel('Normalized
frequency');ylabel('A(\Omega)');
```

Example: Design an FIR high-pass digital filter with the following specifications Ω_p=0.8π rad, Ω_s=0.7π rad，$A_p \le 0.3$dB, $A_s \ge 40$dB

Solution:





M= 62, A_p=0.053dB, A_s=44dB

Example: Given the frequency response of the digital differentiator $H_d(e^{j\Omega})=j\Omega$, $|\Omega|\le\pi$. Design an M=9 linear phase FIR filter using the window method to approximate the ideal digital differentiator.。

Solution:

1. Determine the type of linear phase FIR digital filter based on the design criteria：

   Since $H_d(e^{j\Omega})=j\Omega$ has an odd-symmetric amplitude function and M is odd, choose Type IV.

   Determine the magnitude function of the ideal digital differentiator $A_d(\Omega)$

$$A_d(\Omega)=\begin{cases}\Omega & |\Omega|\le\pi\\ 0 & \text{others}\end{cases}$$

2. Determine the phase of the ideal digital differentiator $\varphi_d(\Omega)$

$$\varphi_d(\Omega)=-0.5M\Omega+p/2$$

Example: Given the frequency response of the digital differentiator $H_d(e^{j\Omega})=j\Omega$, $|\Omega|\le\pi$. Design an $M=9$ linear phase FIR filter using the window method to approximate the ideal digital differentiator.。

Solution:

3. Calculate IDFT to obtain $h_d[k]$

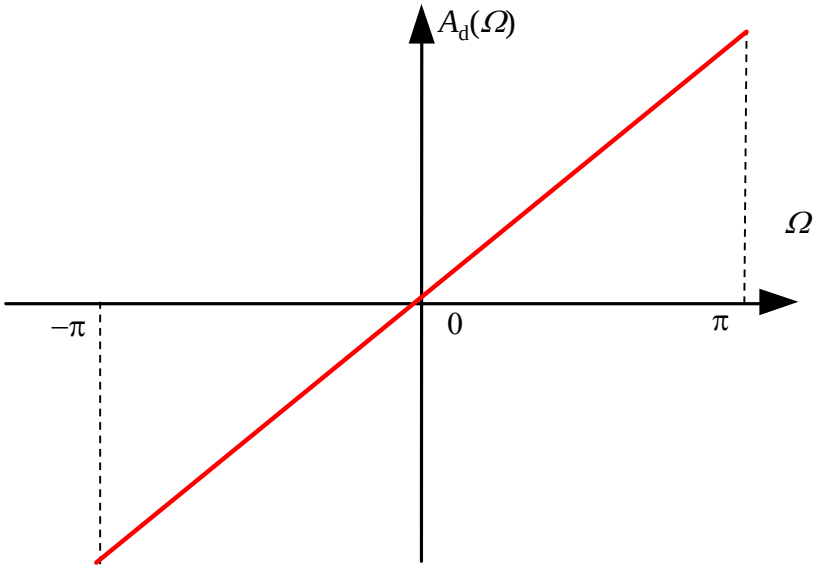
$$h_d[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi} A_d(\Omega)e^{j\varphi_d(\Omega)}e^{jk\Omega}d\Omega$$

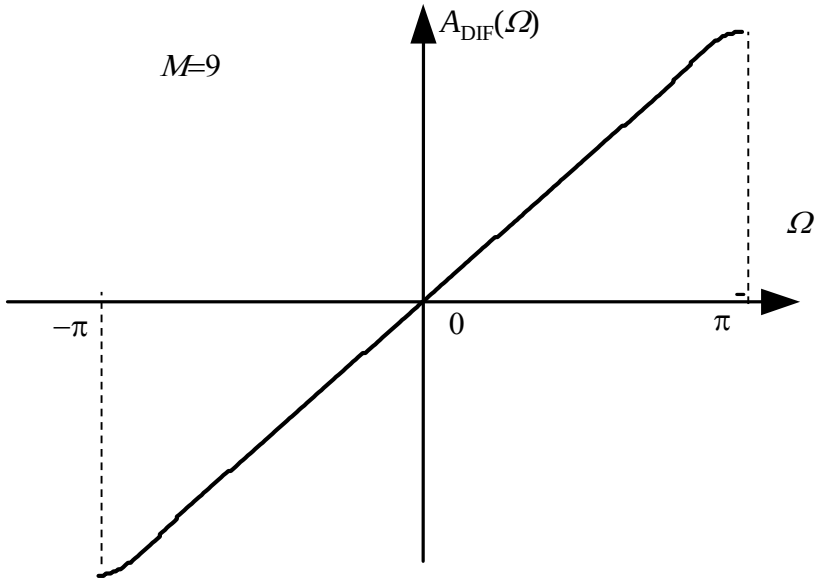
$$=\frac{j}{2\pi}\int_{-\pi}^{\pi}\Omega e^{j\Omega(k-0.5M)}d\Omega=\frac{(-1)^{(k-0.5M+0.5)}}{\pi(k-0.5M)^2}$$

4. Use the rectangular window to truncate $h_d[k]$ to obtain: $h_{DIF}[k]=h_d[k]w_N[k]$

Example: Given the frequency response of the digital differentiator $H_d(e^{j\Omega})=j\Omega$, $|\Omega|\le\pi$. Design an M=9 linear phase FIR filter using the window method to approximate the ideal digital differentiator.。

Solution:





MATLAB implementation of a multi-band filter

[M,Wc,beta,ftype] = kaiserord(f,a,dev); \

Calculate the Kaiser window order and other parameters

input

f: The FIR filter to be designed has *B* frequency bands.

If the FIR filter has 4 frequency bands, they are: $0 \le \Omega \le \pi f_1$ $\pi f_2 \le \Omega \le \pi f_3$ $\pi f_4 \le \Omega \le \pi f_5$ $\pi f_6 \le \Omega \le \pi$

f= [f1 f2 f3 f4 f5 f6];

a: The amplitude values for the *B* frequency bands of the FIR filter to be designed are:

If the amplitude values of the FIR filter in the 4 frequency bands are: a_1 a_2 a_3 a_4

(Passband: 1, Stopband: 0)

a= [a1 a2 a3 a4];

dev: The ripple values for the *B* frequency bands of the FIR filter to be designed are:

If the FIR filter has ripple values in the 4 frequency bands: δ_1 δ_2 δ_3 δ_4

dev= [d1 d2 d3 d4];

MATLAB implementation of a multi-band filter

[M,Wc,beta,ftype] = kaiserord(f,a,dev); Calculate the Kaiser window order and other parameters.

output

M: Estimated order of the FIR filter to be designed.

Wc: The *B* frequency bands of the ideal FIR filter.

beta: Parameter of kaiser

ftype:
- empty (low-pass) :
- 'high' (high-pass) :
- 'stop' (band-stop) :
- 'DC–0' (For the multi-band filter, the first frequency band is a stopband.)
- 'DC–1' (For the multi-band filter, the first frequency band is a passband.)

input

f: The *B* frequency bands of the FIR filter to be designed.

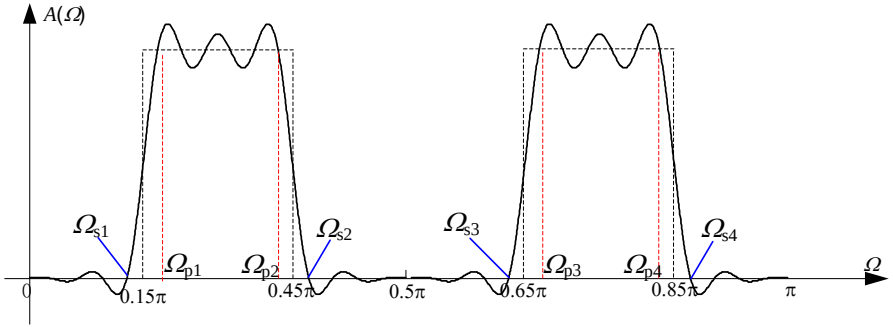
a: The amplitude values for the *B* frequency bands of the FIR filter to be designed. (Passband: 1, Stopband: 0)

dev: The ripple values for the *B* frequency bands of the FIR filter to be designed.

h = fir1(M,Wc,ftype,kaiser(M+1,beta)) Obtain $h[k]$ for the designed FIR filter.

Example: Design a FIR filter with 2 passbands
using the Kaiser window to meet the following specifications:

Ω_{s1}=0.1π rad, Ω_{p1}=0.2π rad, Ω_{p2}=0.4π rad, Ω_{s2}=0.5π rad, Ω_{s3}=0.6π rad,
Ω_{p3}=0.7π rad, Ω_{p4}=0.8π rad, Ω_{s4}=0.9π rad, δ_s=0.008



Frequency band diagram

Key functions: [N,Wc,beta,ftype] = kaiserord(f,a,dev); %Estimate the order *N* and beta
h = fir1(N,Wc,ftype,kaiser(N+1,beta)); %Construct the filter.
mag=freqz(h,[1],omega); % Calculate frequency response

Key parameters: f=[0.1 0.2 0.4 0.5 0.6 0.7 0.8 0.9]; %The frequency bands of the filter
a=[0,1,0,1,0]; %The magnitude values of each frequency band of the filter.
Rs=0.008; dev=Rs*ones(1,length(a)); %Ripple in the stopband.

Example: Design a FIR filter with 2 passbands
using the Kaiser window to meet the following specifications:

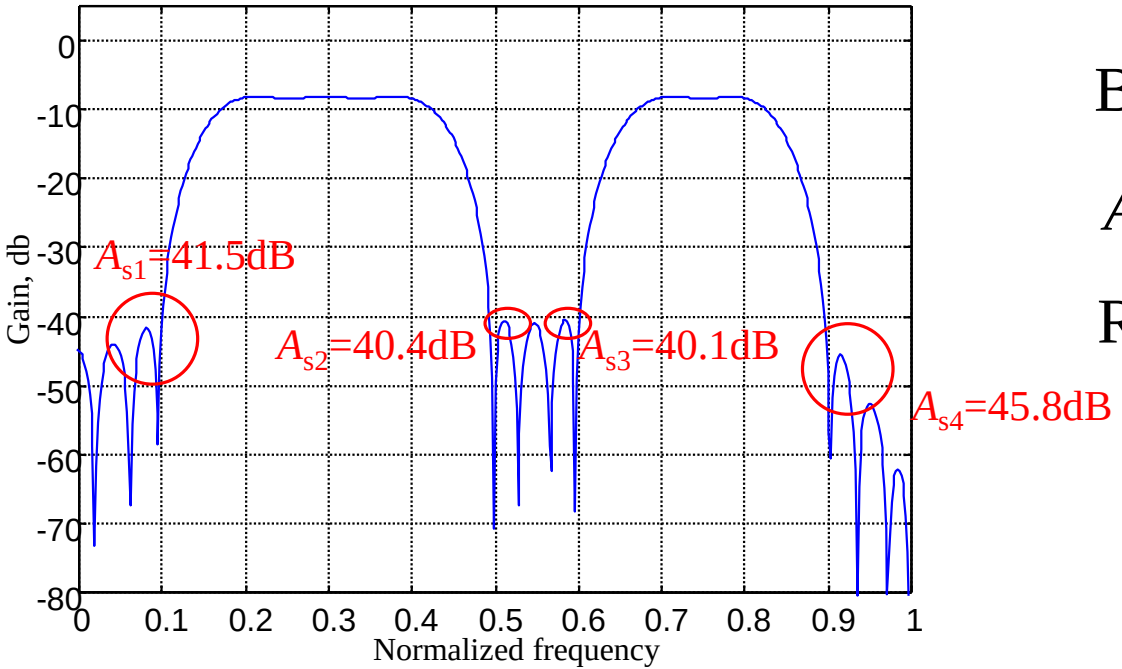
$\Omega_{s1}=0.1\pi$ rad, $\Omega_{p1}=0.2\pi$ rad, $\Omega_{p2}=0.4\pi$ rad, $\Omega_{s2}=0.5\pi$ rad, $\Omega_{s3}=0.6\pi$ rad,
$\Omega_{p3}=0.7\pi$ rad, $\Omega_{p4}=0.8\pi$ rad, $\Omega_{s4}=0.9\pi$ rad, $\delta_s=0.008$

Solution:

```
f=[0.1 0.2 0.4 0.5 0.6 0.7 0.8 0.9]; %The frequency bands of the filter
a=[0,1,0,1,0];Rs=0.008;%The passband amplitude is 1and the stopband is 0.
dev=Rs*ones(1,length(a)); %Ripple in the stopband
[N,Wc,beta,ftype] = kaiserord(f,a,dev); %kaiser estimate the order N and beta
h = fir1(N,Wc,ftype,kaiser(N+1,beta)); %fir1 construct the filter.
omega=linspace(0,pi,512);
mag=freqz(h,[1],omega); % Calculate frequency response
plot(omega/pi,20*log10(abs(mag))); % Calculate frequency response
xlabel('Normalized frequency');
ylabel('Gain, dB');grid;
axis([0 1 -80 5]); %show axis range
```

Example: Design a FIR filter with 2 passbands
using the Kaiser window to meet the following specifications:

$\Omega_{s1}=0.1\pi$ rad, $\Omega_{p1}=0.2\pi$ rad, $\Omega_{p2}=0.4\pi$ rad, $\Omega_{s2}=0.5\pi$ rad, $\Omega_{s3}=0.6\pi$ rad,
$\Omega_{p3}=0.7\pi$ rad, $\Omega_{p4}=0.8\pi$ rad, $\Omega_{s4}=0.9\pi$ rad, $\delta_s=0.008$



By design index:

$$A_s = -20\lg\delta_s = 40\text{dB}$$

Requirements are met

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.